Task 21

Step 3: Enable Detailed Monitoring (EC2)

Go to
👉 Amazon EC2 Console

Steps:

Select your EC2 instance
Click Actions → Monitor and troubleshoot
Click Enable detailed monitoring

👉 This changes:

From 5 min → 1 min metrics

Enable CloudWatch Monitoring

- Create CPU utilization alarm
- Trigger SNS notification
- Monitor ALB metrics
- Enable detailed monitoring for EC2

Step 1: Create SNS Topic (for alerts)

Go to

👉 Amazon SNS Console

Steps:

1. Click **Create topic**
2. Type → **Standard**
3. Name: `cpu-alert-topic`

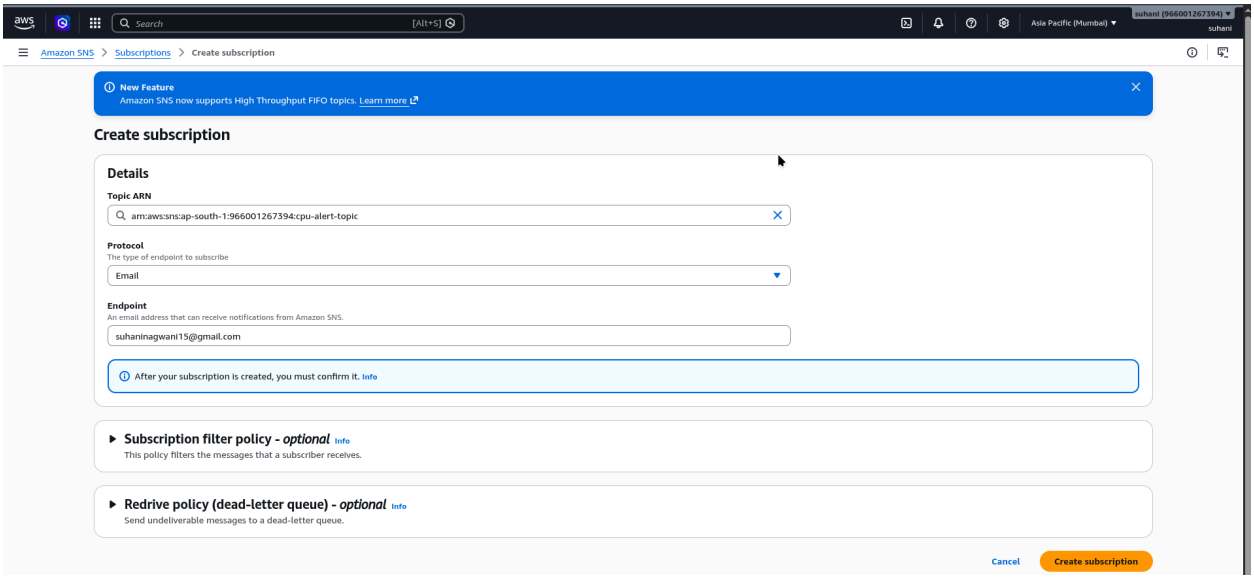
Add Subscription:

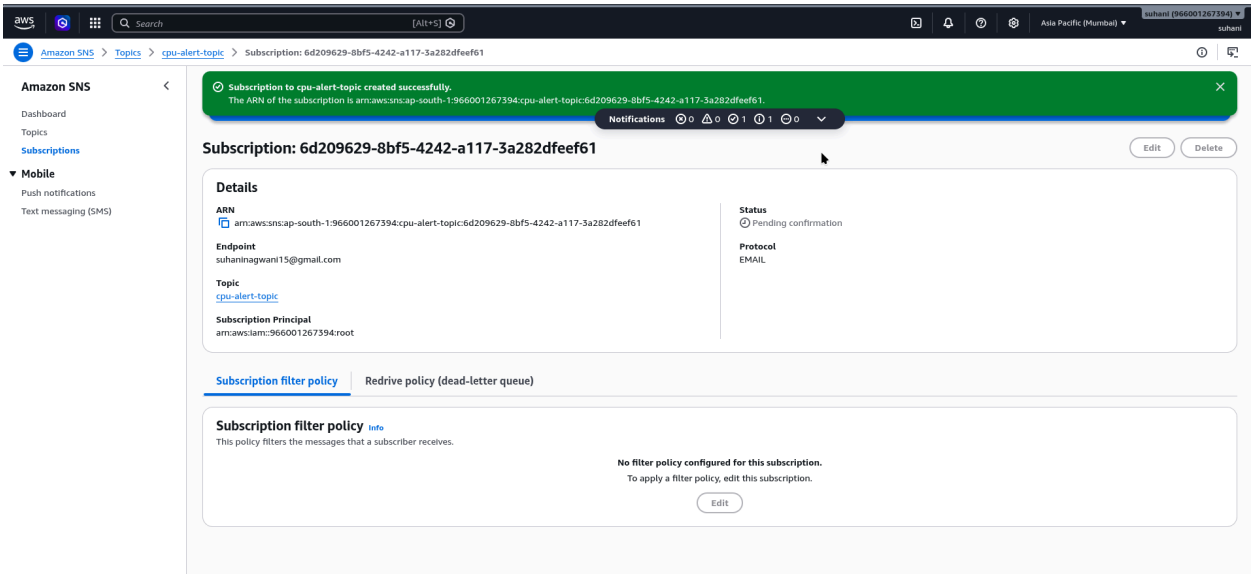
- Protocol: **Email**

- Endpoint: *your email*

👉 Click **Create subscription**

👉 Check email → **Confirm subscription**



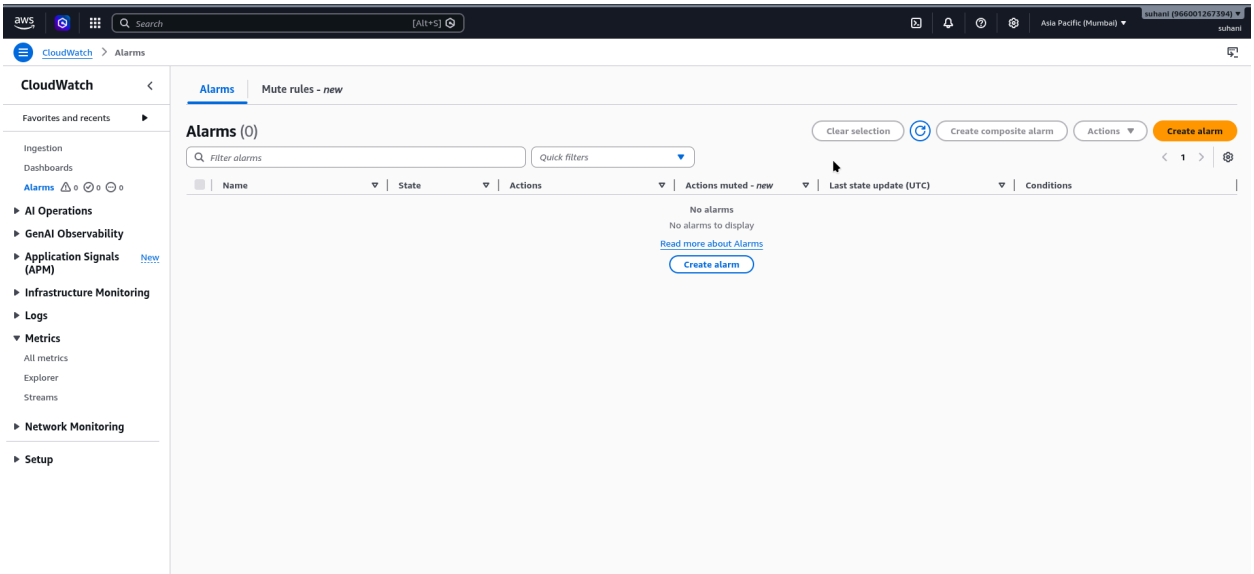


Step 2: Create CPU Utilization Alarm

Go back to CloudWatch:

1. Go to **Alarms** → **Create alarm**

2. Click **Select metric**



Select:

- EC2 → **Per-Instance Metrics**
- Choose:
  - ✔️ `CPUUtilization`

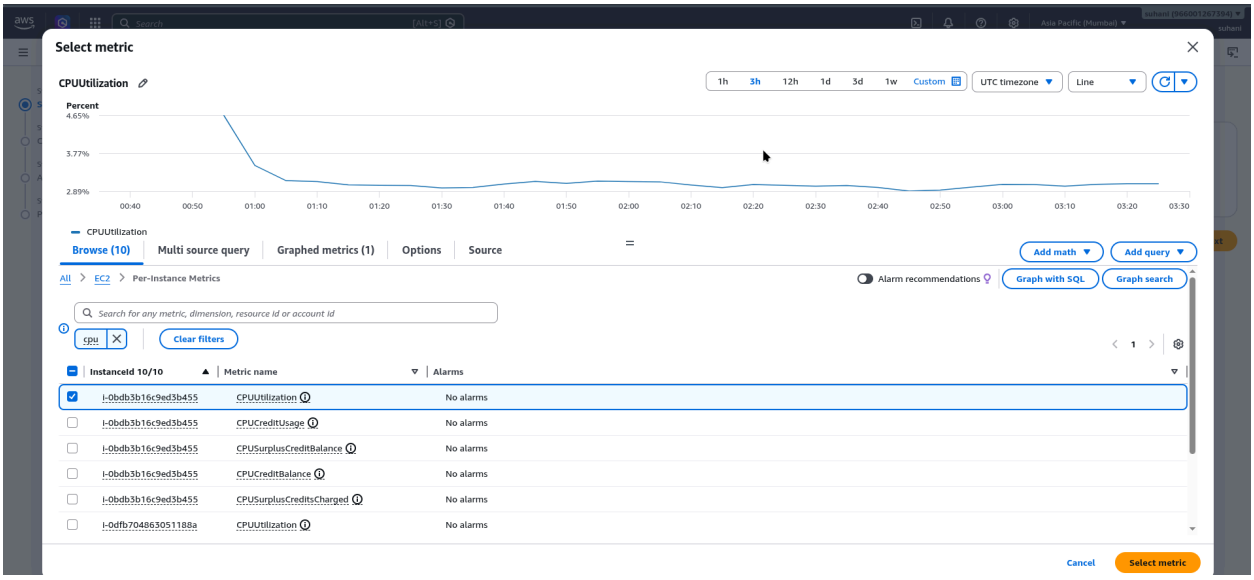
Configure:

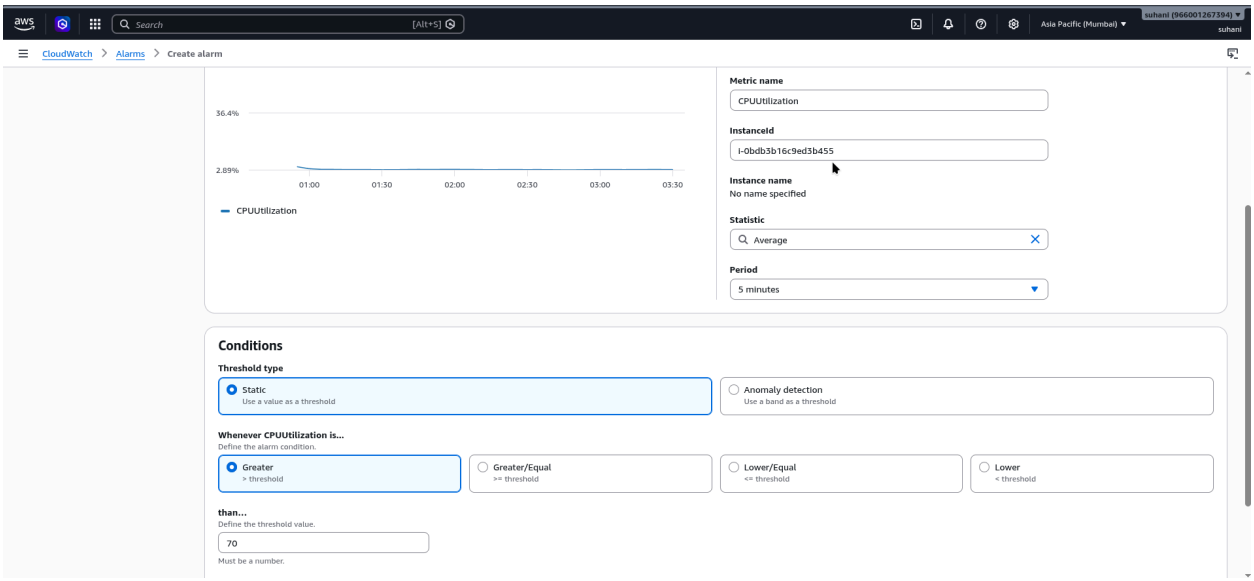
- Threshold type: **Static**
- Condition:
  - **Greater than 70%**
- Period:
  - 5 minutes

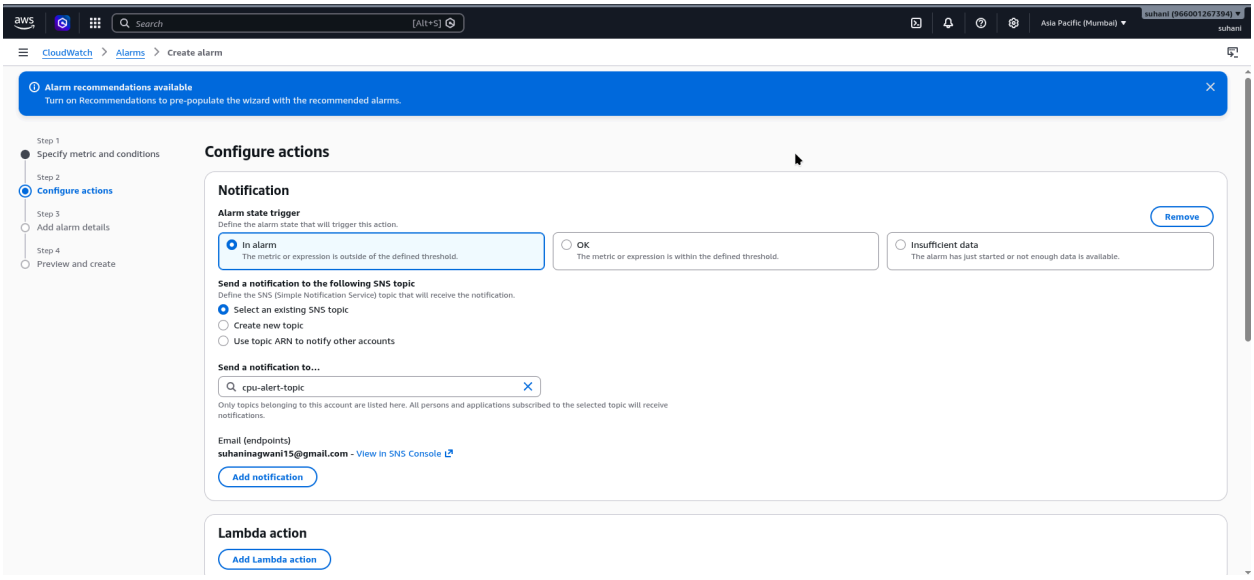
Notification:

- Select SNS topic:
  - `cpu-alert-topic`

👉 Create alarm


Select metric
CPUUtilization
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Line
Percent
4.65%
3.77%
2.89%
00:40 00:50 01:00 01:10 01:20 01:30 01:40 01:50 02:00 02:10 02:20 02:30 02:40 02:50 03:00 03:10 03:20 03:30
CPUUtilization
Browse (10)
Multi source query
Graphed metrics (1)
Options
Source
Add math
Add query
All > EC2 > Per-Instance Metrics
Alarm recommendations
Graph with SQL
Graph search
Search for any metric, dimension, resource id or account id
cpu
Clear filters
InstanceId 10/10
Metric name
Alarms
i-0bdb3b16c9ed3b455 CPUUtilization No alarms
i-0bdb3b16c9ed3b455 CPUCreditUsage No alarms
i-0bdb3b16c9ed3b455 CPUSurplusCreditBalance No alarms
i-0bdb3b16c9ed3b455 CPUCreditBalance No alarms
i-0bdb3b16c9ed3b455 CPUSurplusCreditsCharged No alarms
i-0dfb704863051188a CPUUtilization No alarms
Cancel
Select metric


CloudWatch > Alarms > Create alarm
36.4%
2.89%
01:00 01:30 02:00 02:30 03:00 03:30
CPUUtilization
Metric name
CPUUtilization
InstanceId
i-0bdb3b16c9ed3b455
Instance name
No name specified
Statistic
Average
Period
5 minutes
Conditions
Threshold type
Static
Use a value as a threshold
Anomaly detection
Use a band as a threshold
Whenever CPUUtilization is...
Define the alarm condition.
Greater
> threshold
Greater/Equal
>= threshold
Lower/Equal
<= threshold
Lower
< threshold
than...
Define the threshold value.
70
Must be a number.



Step 3: Enable Detailed Monitoring (EC2)

Go to

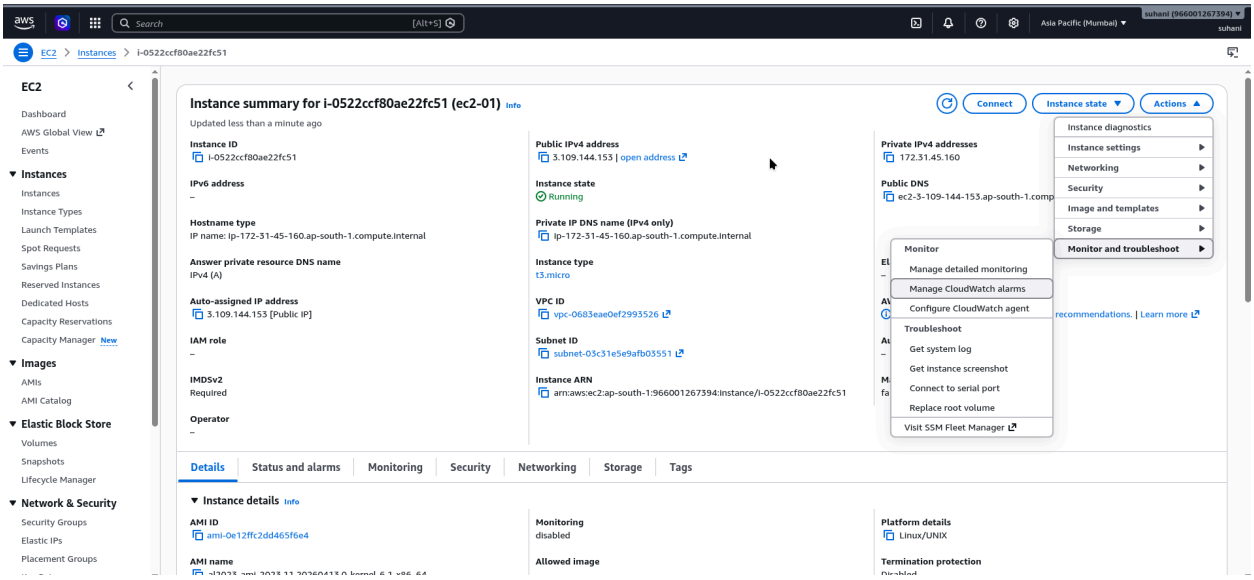
👉 Amazon EC2 Console

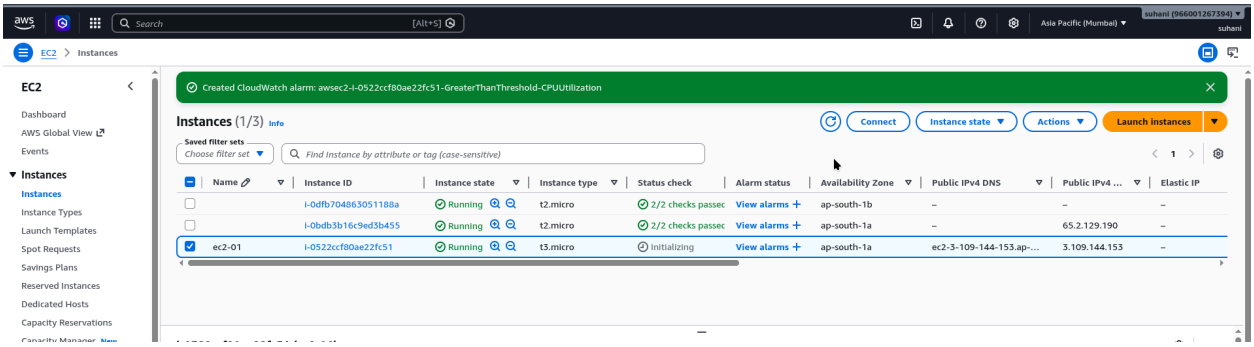
Steps:

1. Select your EC2 instance
2. Click **Actions** → **Monitor and troubleshoot**
3. Click **Enable detailed monitoring**

👉 This changes:

- From **5 min** → **1 min metrics**

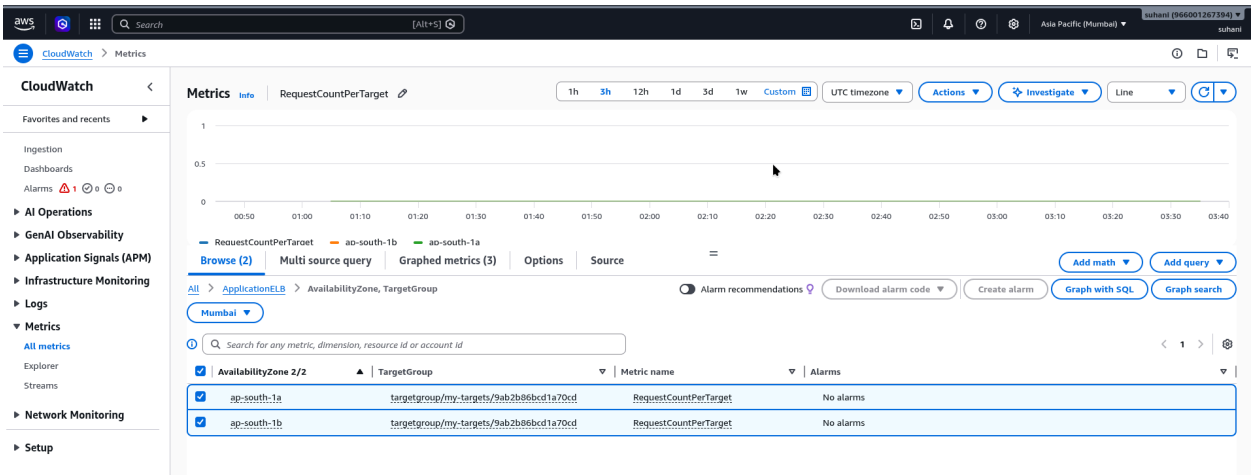




Step 4: Monitor ALB Metrics

Go back to CloudWatch:

1. Go to **Metrics**
2. Click:
    - **ApplicationELB**


aws
Search
[Alt+S]
Asia Pacific (Mumbai)
suhani (966001267394)
suhani
CloudWatch > Metrics
CloudWatch
Favorites and recents
Ingestion
Dashboards
Alarms 1 0 0
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Metrics
All metrics
Explorer
Streams
Network Monitoring
Setup
Metrics Info
RequestCountPerTarget
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Actions
Investigate
Line
1
0.5
0
00:50 01:00 01:10 01:20 01:30 01:40 01:50 02:00 02:10 02:20 02:30 02:40 02:50 03:00 03:10 03:20 03:30 03:40
RequestCountPerTarget
ap-south-1b
ap-south-1a
Browse (2)
Multi source query
Graphed metrics (3)
Options
Source
Add math
Add query
All > ApplicationELB > AvailabilityZone, TargetGroup
Alarm recommendations
Download alarm code
Create alarm
Graph with SQL
Graph search
Mumbai
Search for any metric, dimension, resource id or account id
1
AvailabilityZone 2/2
TargetGroup
Metric name
Alarms
ap-south-1a
targetgroup/my-targets/9ab2b86bcd1a70cd
RequestCountPerTarget
No alarms
ap-south-1b
targetgroup/my-targets/9ab2b86bcd1a70cd
RequestCountPerTarget
No alarms